

Assignment 9: SQL Date & Time Functions

Overview:

Practice SQL date and time functions by calculating ages, experience, and extracting components. Learn to filter dates dynamically.

1. Create and Populate Table

```
```sql
CREATE TABLE Employees (
 EmpID INT PRIMARY KEY,
 Name VARCHAR(50),
 DateOfBirth DATE,
 JoinDate DATE
);
INSERT INTO Employees (EmpID, Name, DateOfBirth, JoinDate) VALUES
(1, 'Alice Johnson', '1985-05-20', '2010-06-15'),
(2, 'Bob Smith', '1990-08-10', '2015-09-01'),
(3, 'Charlie Brown', '1988-03-25', '2012-11-12'),
(4, 'Diana Prince', '1992-01-30', '2017-07-08'),
(5, 'Eve Adams', '1987-12-05', '2013-03-20');
```
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot_1**) and save it in the same **PDF/Doc/Word** file.

2. SQL Query Tasks

1. **a) Current Date & Time:** Display the current system date and time.

SQL:

```
SELECT CURRENT_TIMESTAMP;
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot_2a**) and save it in the same **PDF/Doc/Word** file.

2. **b) Calculate Age:** Compute each employee's age in years based on DateOfBirth.

SQL:

```
SELECT Name, FLOOR(DATEDIFF(CURRENT_DATE, DateOfBirth)/365.25) AS Age
FROM Employees;
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot_2b**) and save it in the same **PDF/Doc/Word** file.

3. **c) Calculate Experience:** Compute total years of service since JoinDate.

SQL:

```
SELECT Name, FLOOR(DATEDIFF(CURRENT_DATE, JoinDate)/365.25) AS  
YearsExperience FROM Employees;
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot_2c**) and save it in the same **PDF/Doc/Word** file.

4. **d) Extract DOB Components:** Extract year, month, and day from DateOfBirth.

SQL:

```
SELECT Name, YEAR(DateOfBirth) AS BirthYear, MONTH(DateOfBirth) AS BirthMonth,  
DAY(DateOfBirth) AS BirthDay FROM Employees;
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot_2d**) and save it in the same **PDF/Doc/Word** file.

5. **e) Born in August:** List employees born in the month of August.

SQL:

```
SELECT Name FROM Employees WHERE MONTH(DateOfBirth) = 8;
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot_2e**) and save it in the same **PDF/Doc/Word** file.

6. **f) Upcoming Birthdays:** Display names of employees whose birthdays fall in the next 30 days.

SQL:

```
SELECT Name FROM Employees WHERE  
DATE_FORMAT(DATE_ADD(DateOfBirth, INTERVAL  
YEAR(CURRENT_DATE)-YEAR(DateOfBirth) YEAR), '%m-%d')  
BETWEEN DATE_FORMAT(CURRENT_DATE, '%m-%d') AND  
DATE_FORMAT(DATE_ADD(CURRENT_DATE, INTERVAL 30 DAY), '%m-%d');
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output (**as screenshot_2f**) and save it in the same **PDF/Doc/Word** file.

Submission Guidelines:

Create One Main Folder (named [Assignment9_YourName](#)) which includes -

- One **Doc/PDF/Word** file containing all the Queries and the screenshots of each part.
- Put your SQL file as **Assignment9_YourName.sql**

Convert the folder into ZIP format.

Upload the ZIP folder on the submission portal.

Watch this to learn how to create ZIP file

https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact the Tutedude mentor.